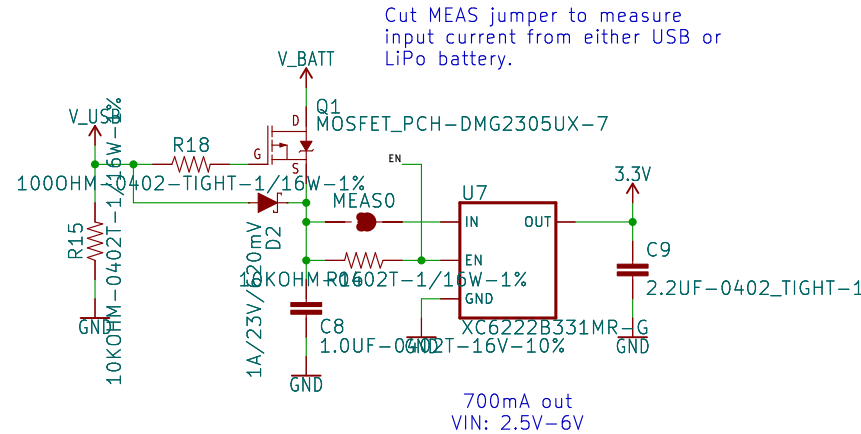


[illegible]

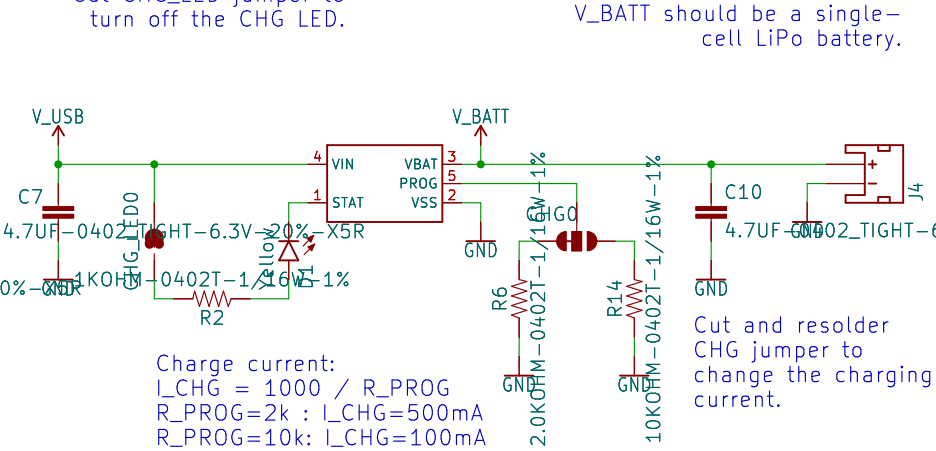
Lithium-Polymer Battery Charger (1-cell)

Cut CHG_LED jumper to turn off the CHG LED.

V_BATT should be a single-cell LiPo battery.

Charge current:
I_CHG = 1000 / R_PROG
R_PROG=2k : I_CHG=500mA
R_PROG=10k: I_CHG=100mA

Cut and resolder CHG jumper to change the charging current.

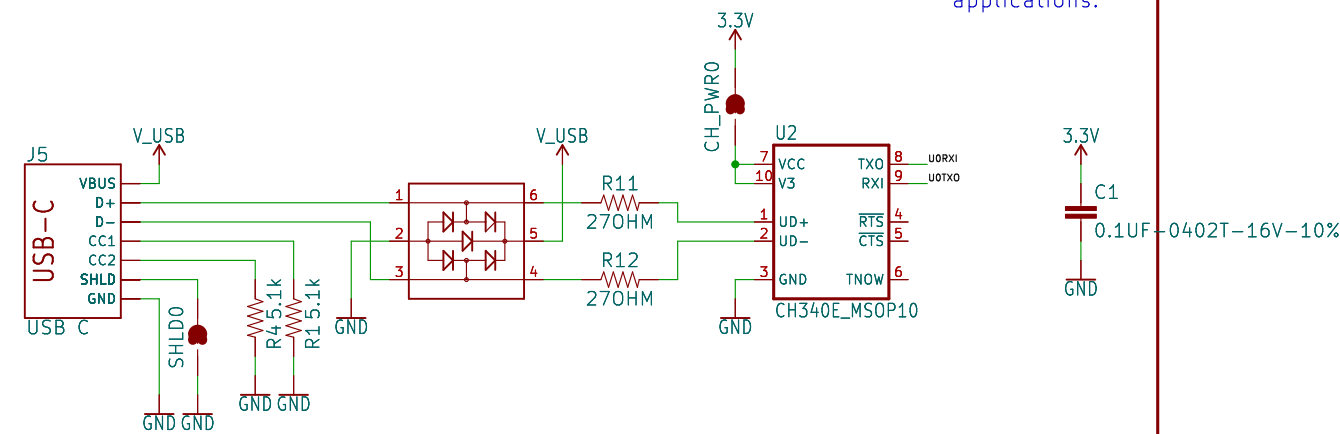


20%-X5R

USB-C Firmware Upload

Cut CH_PWR jumper to remove CH340 from the 3.3V rail for low-power applications.

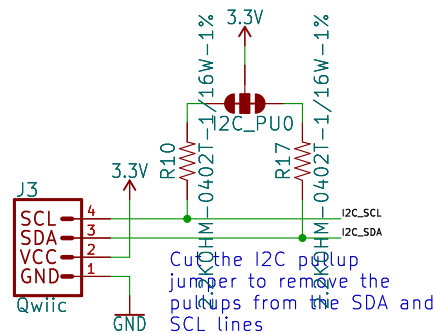
Diagram illustrating the USB-C Firmware Upload circuit. The circuit includes a USB-C connector (J5) connected to a USB-C to UART bridge (U2, CH340E_M50P10) via a diode bridge and resistors (R11, R12). The bridge is connected to a 3.3V rail (CH_PWR0) which has a jumper (C1) to GND. The 3.3V rail is also connected to a 3.3V source. The USB-C connector pins are labeled: VBUS, D+, D-, CC1, CC2, SHLD, GND. The bridge pins are labeled: V_USB, GND, TXO, RXI, U0RXI, U0TXO, RTS, CTS, TNOW. The 3.3V rail is labeled: CH_PWR0, 3.3V, GND. The capacitor C1 is labeled: 0.1uF, 0402T-16V-10%.



Qwiic Connector

The diagram illustrates the Qwiic connector circuit. A 3.3V supply is connected to the VCC pin (pin 2) of the connector. The GND pin (pin 1) is connected to ground. The SCL pin (pin 4) is connected to the 3.3V supply through a pull-up resistor R10 (0.402Ω, 1/16W, 1%). The SDA pin (pin 3) is connected to the 3.3V supply through a pull-up resistor R17 (0.402Ω, 1/16W, 1%). A 2C_PU0 jumper is connected between the SCL and SDA pins. The I2C_SCL and I2C_SDA labels indicate the connection points for the I2C lines.

Cut the I2C pullup jumper to remove the pullups from the SDA and SCL lines



DA16200 Module

The RTC_PWR_KEY pin enables the RTC block of the DA16200. Keep this pin connected to 3.3V for normal operation. Flip the switch to connect to GND and enter sleep mode 1. Refer to the datasheet for more information.

Cut DA_MEAS jumper to measure the current draw from just the DA16200 module.

Debug Header

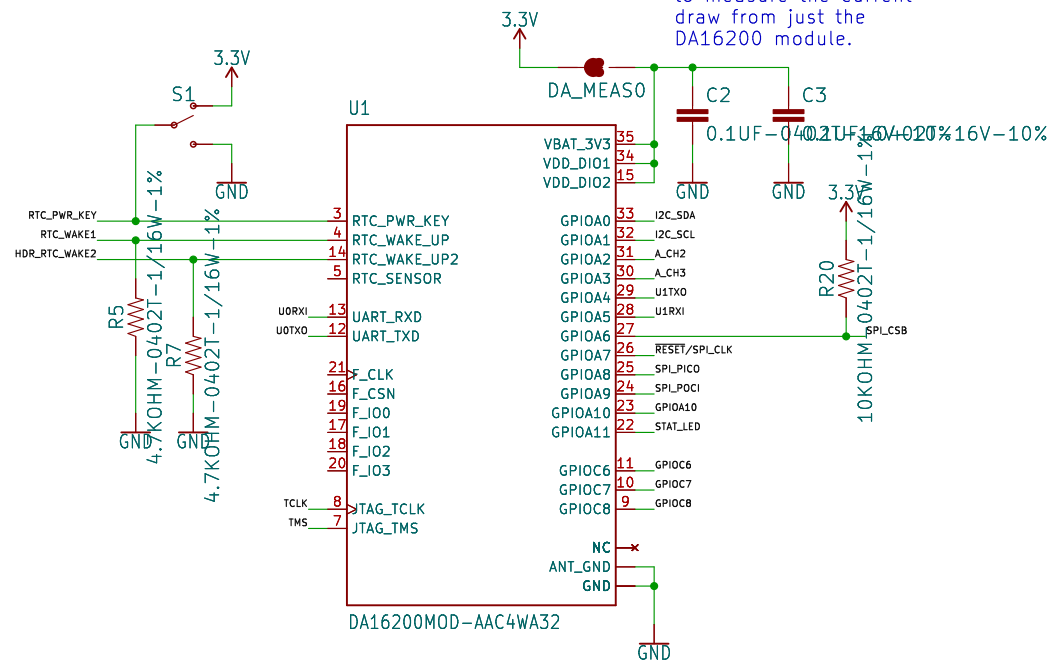
Power LED

Cut LED jumper to remove LED from power.

Status LED

more information.

Cut DA_MEAS0 jumper to measure the current draw from just the DA16200 module.

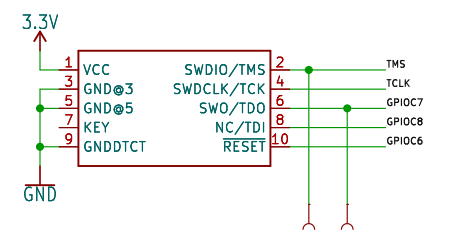


Pinout diagram of the STM32F103C8T6 microcontroller:

- Pin 1: VCC (3.3V)
- Pin 2: SWDIO/TMS
- Pin 3: GND@3
- Pin 4: SWDCLK/TCK
- Pin 5: GND@5
- Pin 6: SWO/TDO
- Pin 7: KEY
- Pin 8: NC/TDI
- Pin 9: GND@TCT
- Pin 10: RESET

Additional connections shown:

- TMS (connected to Pin 2)
- TCLK (connected to Pin 4)
- GPIOD7 (connected to Pin 6)
- GPIOD8 (connected to Pin 8)
- GPIOD6 (connected to Pin 10)



Power LED Status LED

3.3V

LED0

D3

R8

0402T-1/16W

1K0

5V

STATUS LED

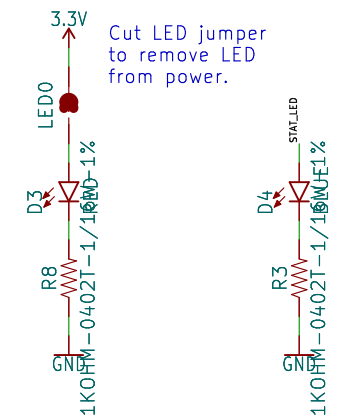
D4

R3

0402T-1/16W

1K0

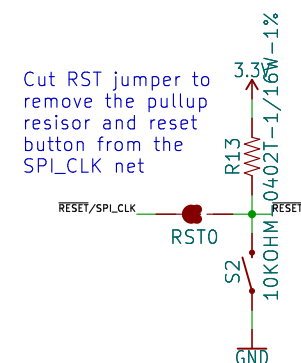
Cut LED jumper to remove LED from power.



Reset Button

Cut RST jumper to remove the pullup resistor and reset button from the SPI_CLK net

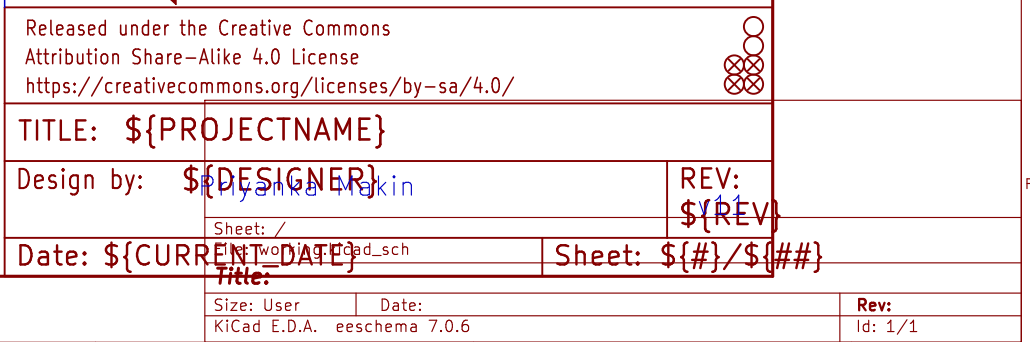
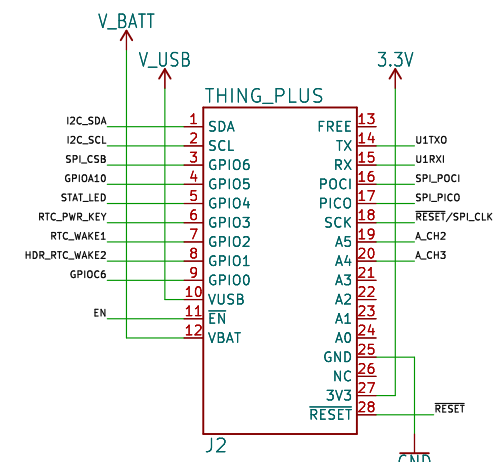
```
graph LR
    SPI_CLK[RESET/SPI_CLK] --- RSTO((RSTO))
    RSTO --- J1(( ))
    J1 --- R13[R13 0402T-1/16W-1%]
    R13 --- V33[3.3V]
    J1 --- S2[S2 10KOHM]
    S2 --- GND[GND]
```



Thing Plus

The diagram illustrates the pinout for the Thing Plus module. The pins are numbered 1 through 28. The connections are as follows:

- Left Side (Pins 1-12):**
 - 1: I2C_SDA
 - 2: I2C_SCL
 - 3: SCL
 - 4: GPIOD5
 - 5: GPIOD4
 - 6: GPIOD3
 - 7: GPIOD2
 - 8: GPIOD1
 - 9: GPIOD0
 - 10: VUSB
 - 11: EN
 - 12: VBAT
- Right Side (Pins 13-28):**
 - 13: FREE
 - 14: TX
 - 15: RX
 - 16: POC
 - 17: PICO
 - 18: SCK
 - 19: A5
 - 20: A4
 - 21: A3
 - 22: A2
 - 23: A1
 - 24: A0
 - 25: GND
 - 26: NC
 - 27: 3V3
 - 28: RESET
- Power and Ground Connections:**
 - V_BATT (Pin 1)
 - V_USB (Pin 2)
 - 3.3V (Pin 27)
 - GND (Pin 25)
- Other Connections:**
 - RESET (Pin 28)
 - TX (Pin 14)
 - RX (Pin 15)
 - POC (Pin 16)
 - PICO (Pin 17)
 - SCK (Pin 18)
 - A5 (Pin 19)
 - A4 (Pin 20)
 - A3 (Pin 21)
 - A2 (Pin 22)
 - A1 (Pin 23)
 - A0 (Pin 24)
 - NC (Pin 26)
 - 3V3 (Pin 27)



TITLE: \${PROJECTNAME}			
Design by: \${DESIGNER}	REV: \${REV}		
Sheet: /			
Date: \${CURRENT_DATE}	Sheet: \${#}/\${##}		
Size: User		Date:	Rev:
KiCad E.D.A. eschema 7.0.6			Id: 1/1